

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE – RAIGAD -402 103**

Supplementary Semester Examination – January 2023

Branch: Mechanical Engineering

Sem.:- IV

Marks: 100

Subject with Subject Code:- Manufacturing Processes –I (BTMEC401)

Date:-

Time:- 3 Hr.

Instructions to the Students

1. Each question carries 12 marks. All questions are compulsory.
2. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
3. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly

1		Answer Any Two of the following:	Level/CO	Marks
	A	Explain Investment Casting Process in detail with a suitable sketch.	CO1/Recall	6
	B	Explain any Five casting defects in brief. What are their causes and how they are eliminated?	CO1/Recall	6
	C	Dictate any four required properties of molding sand. Draw part gating system showing different gating elements	CO1/Understand	6
2		Answer Any Two of the following:		
	A	Explain typical arrangement of role and mills with neat sketches.	CO2/Recall	6
	B	What are the types of forging drop hammers? Explain any one power drop hammer with a neat sketch.	CO2/Recall	6
	C	What are the disadvantages of forging processes? Explain why parts produced by forging is preferred when compared to other machining and welding processes?	CO2/Understand	6
3		Answer Any Two of the following:		
	A	What is the basic principle of metal drawing? Explain the process of deep drawing used for manufacturing of a cup. Show with a neat & simple sketch.	CO3/Recall	6
	B	Differentiate between direct extrusion & indirect extrusion process. Explain in brief the limitations of metal extrusion process.	CO3/Understand	6
	C	Explain the following basic sheet metal forming processes. A) Blanking B) Bending C) Lancing	CO3/Recall	6

4		Answer Any Two of the following:		
	A	What are four different types of MIG welding processes? Explain any one of the above technique with a simple diagram.	CO4/Recall	6
	B	Classify different regions of Oxy acetylene flame and with the help of sketch explain their characteristics.	CO4/Recall	6
	C	What are the types of Brazing processes? Write the advantages and limitations of brazing process.	CO4/Recall	6
5		Answer Any Two of the following:		
	A	Classify Lathe machines? What are the main components of a Lathe machine?	CO5/Recall	6
	B	Name the various gear finishing methods. Explain in brief the Gear shaving process with a neat Sketch.	CO5/Understand	6
	C	Differentiate between Down milling and Up milling. What is mean by straddle milling?	CO5/Understand	6

Note: Please remove the grid before printing the question paper.